



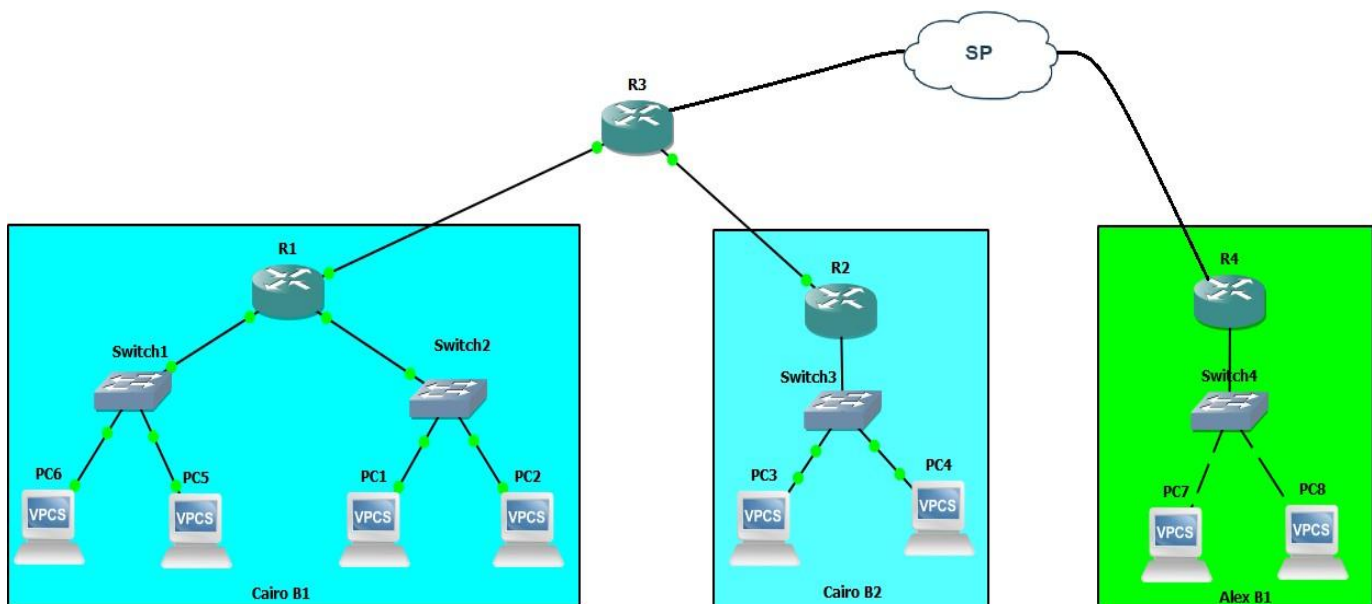
Site Design

Company X is an autonomous system (AS) with multiple branches located in Cairo and Alexandria. The assignment is to design and implement a network that connects these sites' local area networks (LANs), ensuring reliable communication and efficient data transfer between them.

Objective:

- Learn how to design, implement, and configure small C large network
 - IP scheme.
 - Switching.
 - Routing (EIGRP C BGP).

Branch Name	Building Number	Department Name	Number of users	Network
Cairo	Building 1 (B1)	Sales (Sw1)	2 PCs	20.1.1.0/24
		HR (Sw2)	2 PCs	30.1.1.0/24
	Building 2 (B2)	IT (SW3)	2 PCs	40.1.1.0/24
Alex	Building 1 (B1)	Operation (SW4)	2 PCs	50.1.1.0/24





-You have been hired as a network engineer to plan, design, implement, and troubleshoot the network using IPv4

Project Brief:

1. **Network IP Assignment:** Provide a separate Network IP for each department and further subnet the network for each department as specified in the table above.
2. **FLSM Strategy:** Fixed Length Subnet Masking (FLSM) strategy using 256 hosts for each network.
3. **Routing Protocol (EIGRP):** Implement the EIGRP (Enhanced Interior Gateway Routing Protocol) for internal routing between Router RW1 and Router RW3, and between Router RW2 and Router RW3, to ensure that users in Building 1 (B1) can communicate with users in Building 2 (B2).
4. **Routing Protocol (BGP):** Implement BGP (Border Gateway Protocol) for external routing between Router RW3 and Router RW4 to enable communication between the Cairo and Alexandria branches.
5. **PC Assignment:** Allocate two PCs per department.
6. **Network Reachability:** Ensure that all PCs on the network are reachable from any other PC.

Notes:

- 1- **Simulation Platform:** The assignment will be implemented using the GNS3 simulator.
- 2- **Team Size:** The assignment is to be completed in a group of up to two members.
- 3- **Report Submission:** A detailed report must be submitted, outlining the network plan, design, and IP addressing scheme.